

Azure Content Understanding

Global AI community

12 mars 2026

Présentation du service Azure Content Understanding



**global AI
community**
Virtuelle Francophone

Présentation du service Azure Content Understanding

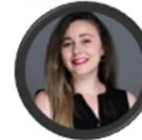
Azure Content Understanding est un service Microsoft Foundry qui **automatise l'extraction et la structuration de contenu multimodal (documents, images, vidéos, fichiers audios) sans labélisation complexe.**

Une présentation de ce service à l'aide de démonstrations sera réalisée lors de ce webinaire.



Serge RETKOWSKY

France AI Global Black Belt
@Microsoft



Randa MEZRAG

Consultante DATA & IA
@Finelog-Biseum

LinkedIn Live

jeudi

12/03/2026

à partir de 13h

<https://www.linkedin.com/events/7428034373116928000>

Agenda

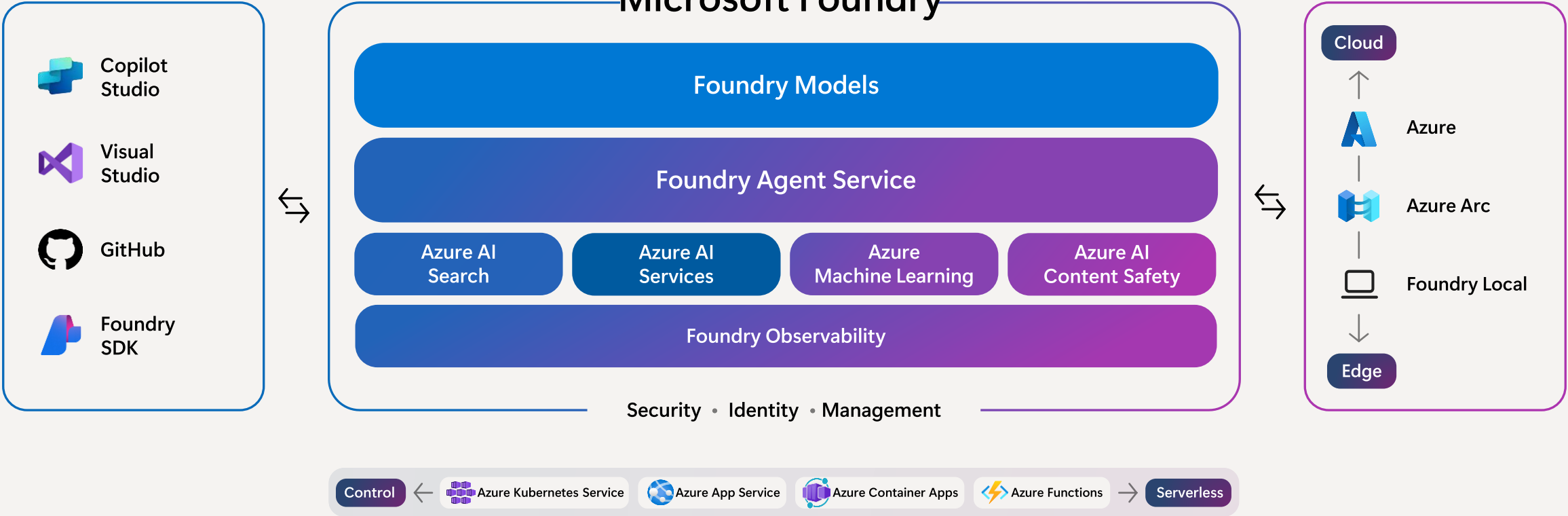
- **Présentation Azure Content Understanding**
- Démonstrations
- Azure Document Intelligence vs Azure Content Understanding
- Ressources et documentation



Microsoft Foundry



Microsoft Foundry



Foundry Tools – Azure services

Content Understanding

- Multi-file processing
- Multi-modal processing
- Prebuilt Analyzers
- Documents
- Images
- Video
- Audio

Document Intelligence

- Document Analysis
- Prebuilt Models
- Custom
- Add-on Capabilities

Language

- Entity extraction (PII/PHI)
- Summarization
- Text analytics
- Sentiment mining
- Conversational AI

Speech

- Speech to text
- Text to speech
- Custom Neural Voice
- Speech Translation
- Avatar
- Embedded speech

Translator

- Text translation
- Document translation
- Speech translation
- Translator Pro
- Embedded translation

Vision

- OCR
- Facial Recognition

Foundry Tools Availability Options



Microsoft
Public Cloud



Microsoft
Government Cloud



Connected
Containers



Disconnected
Containers

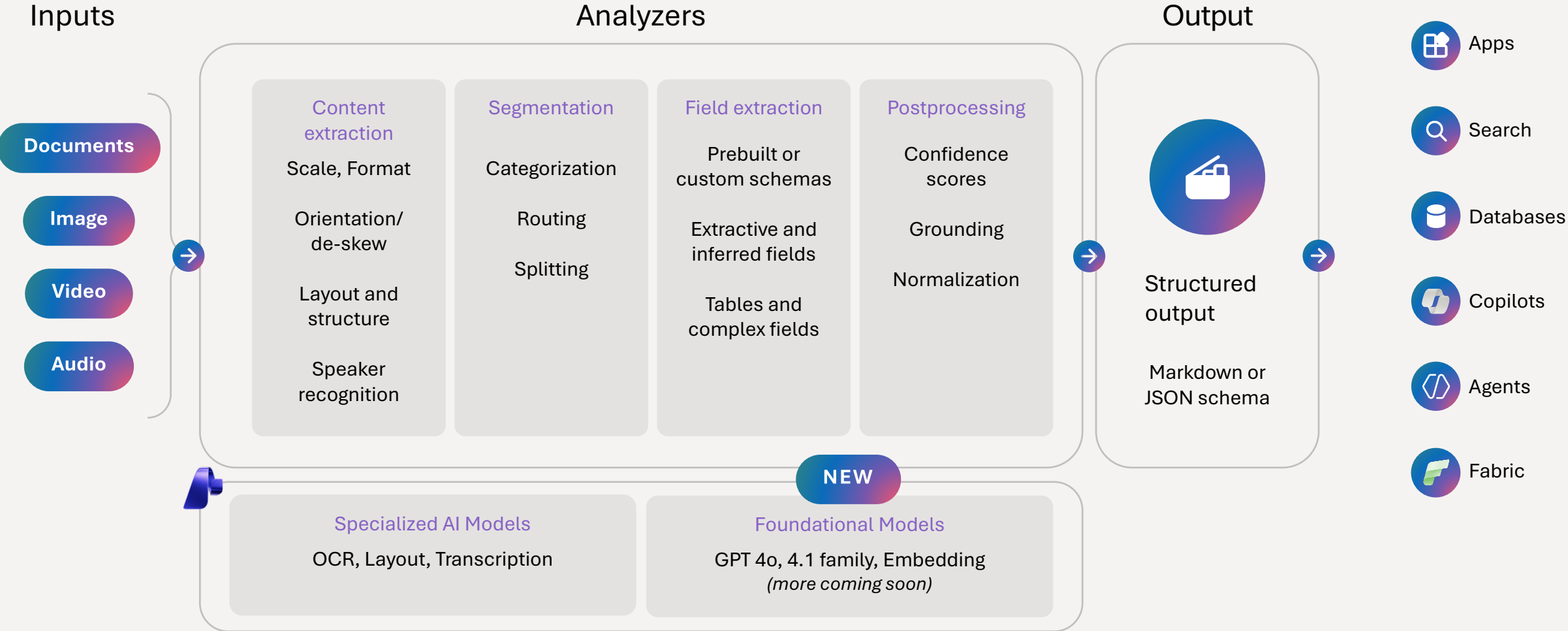
Regulatory
Compliance



HITRUST



Content Understanding framework



Content Understanding is now a Generally Available (GA) service with the release of the 2025-11-01 API version.

Content Understanding

Microsoft Foundry / demos-resource-project

Search with AI (Ctrl + K)

New Foundry

Home

Discover

Build

Operate

Docs

Agents

Workflows

Models

Fine-tune

Tools

Knowledge

Data

Evaluations

Guardrails

Models

Deployments AI Services Batch Jobs

Name	Version	Type
☐ Azure Speech - Voice Live	N/A	Speech
☐ Azure Speech - Speech to Text	N/A	Speech
☐ Azure Speech - Text to Speech	N/A	Speech
☐ Azure Speech - Text to Speech Avatar	N/A	Speech
☐ Azure Translator - Text translation	N/A	Translation
☐ Azure Translator - Document translation	N/A	Translation
☐ Azure Language - Language detection	N/A	Language
☐ Azure Language - Text PII redaction	N/A	Language
☐ Azure Content Understanding - Read	N/A	Content Understanding
☐ Azure Content Understanding - Layout	N/A	Content Understanding

1-10 of 10

< Prev Next >

Content Understanding

Microsoft Foundry / demos-resource-project

Search with AI (Ctrl + K)

New Foundry Home

Agents

Workflows

Models

Fine-tune

Tools

Knowledge

Data

Evaluations

Guardrails

← Azure Content Understanding - Layout

Drag & drop files here or Browse for files

Sample

layout_financi...

Sample

layout_formulas.png

Sample

layout_pageobject.pdf

ticket de caisse.jpg

Run analysis



1 of 1

Content Result

Markdown Paragraphs Tables

Current table

1 of 2

NOM	QT	PT
MENU FIL SAISONS	3	84.00
HOSPITALET ROUGE	1	24.00
DEMI SANPELLEGRI	1	2.80
Sous total :		110.80

	HT	TVA	TTC
Tva 20.0%:	20.00	4.00	24.00
Tva 10.0%:	78.91	7.89	86.80

Total du: 110.80

relais.pithiviers@shfrance.com
Sarl Rouen hotel
1e Relais Saint Georges
au capital de 125.920 Euros
RCS 395 288 004 00074 NAF 5510Z
Intracom FR 833 952 88 004
Tel : 02 38 30 40 25
Fax : 02 38 30 09 05

Content Understanding Extraction Capabilities



Documents

- OCR – printed or handwritten text from 100+ languages
- Content elements (selection mark, check box, radio buttons, barcodes, formulas, image elements)
- Layout elements (pages, paragraphs, lines, sections)
- Tabular data (including complex multi-page layouts)
- Prebuilt Analyzers (including RAG-ready extractions)
- Multi-document input (Pro mode)
- Multi-file reasoning (Pro mode)
- Reference data (Pro mode)
- Classification & splitting



Image

- Generate titles, descriptions, and summaries
- Classify elements
- Prebuilt Analyzer



Audio

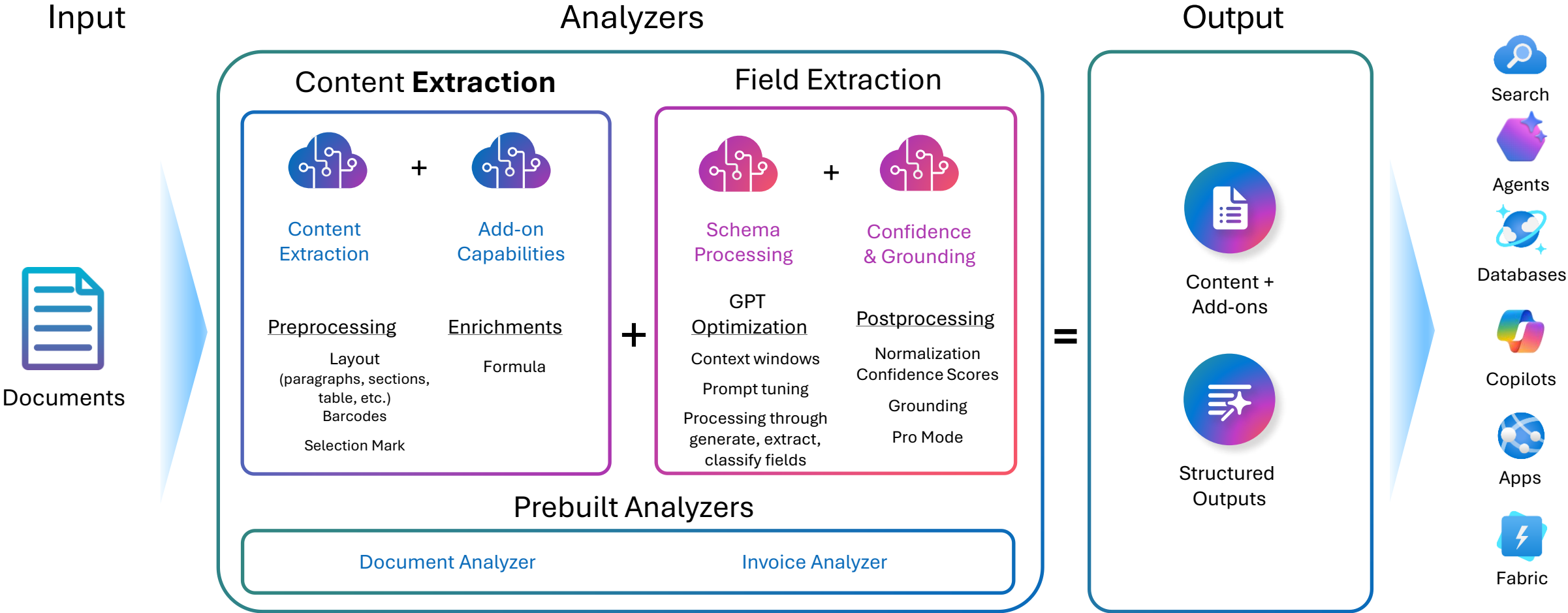
- Transcription
 - Diarization
 - Multi-lingual Transcription
 - Language Detection
 - Multi-Speaker role detection
- Generate conversation summaries and transcripts
- Determine sentiment
- Prebuilt Analyzers



Video

- Transcription
 - Diarization
 - Multi-lingual Transcription
 - Language Detection
 - Key frame extraction
 - Shot detection
- Segmentation (whole-video, automatic, custom)
- Face Detection and Recognition (gated)
- Generate summaries
- Determine sentiment
- Prebuilt Analyzers
 - RAG-ready extraction (prebuilt video indexing)

Document Capabilities



Audio Capabilities

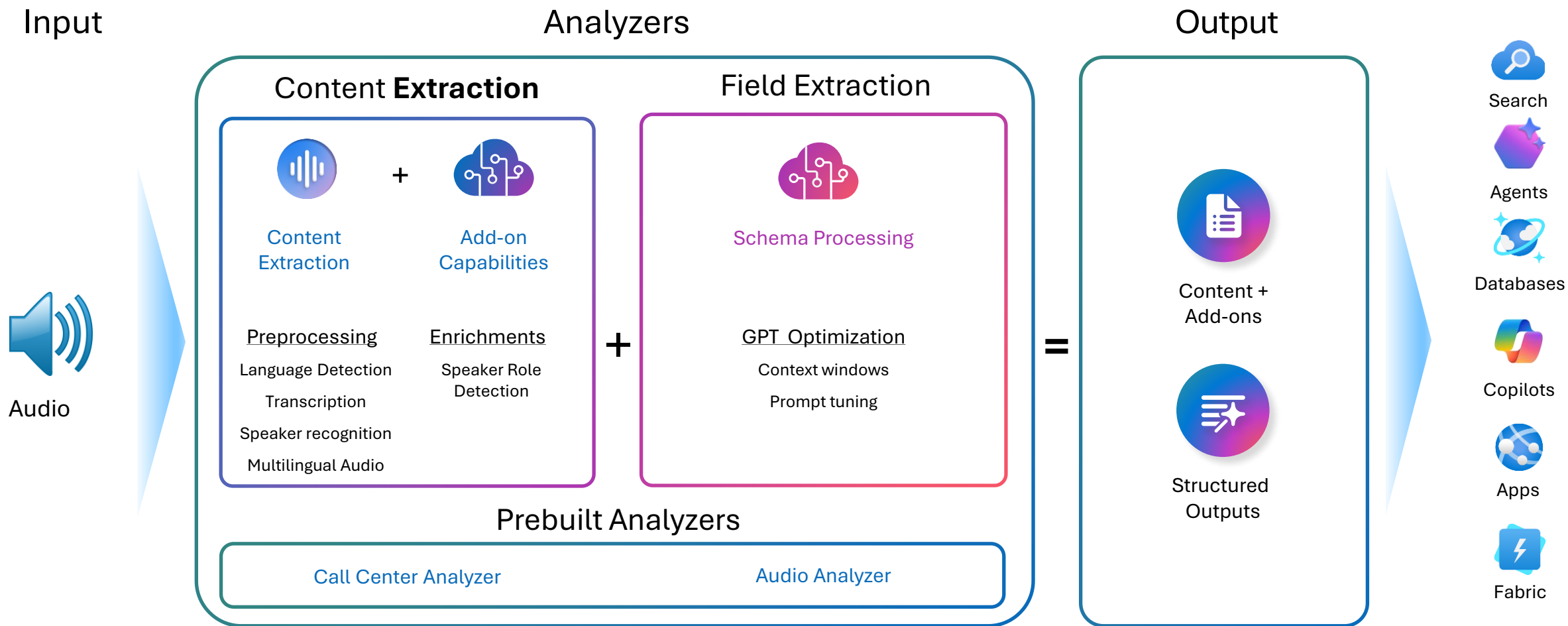
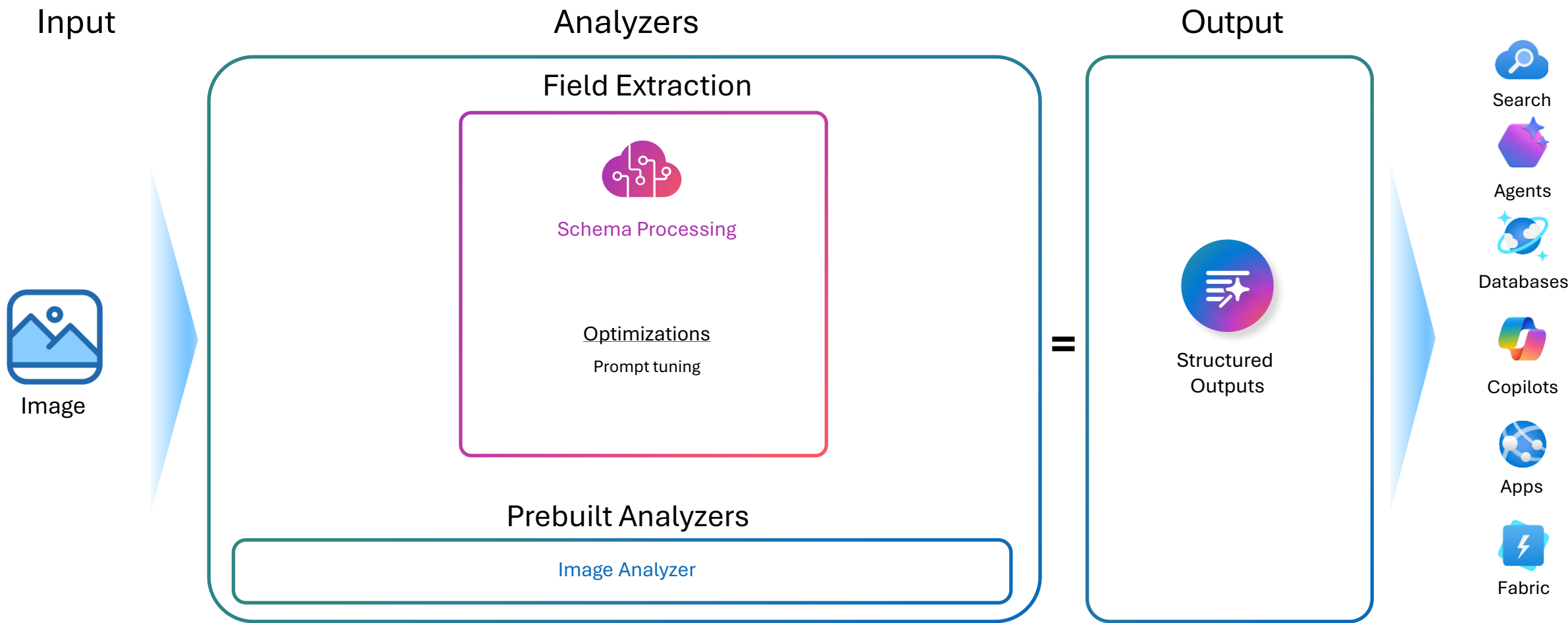
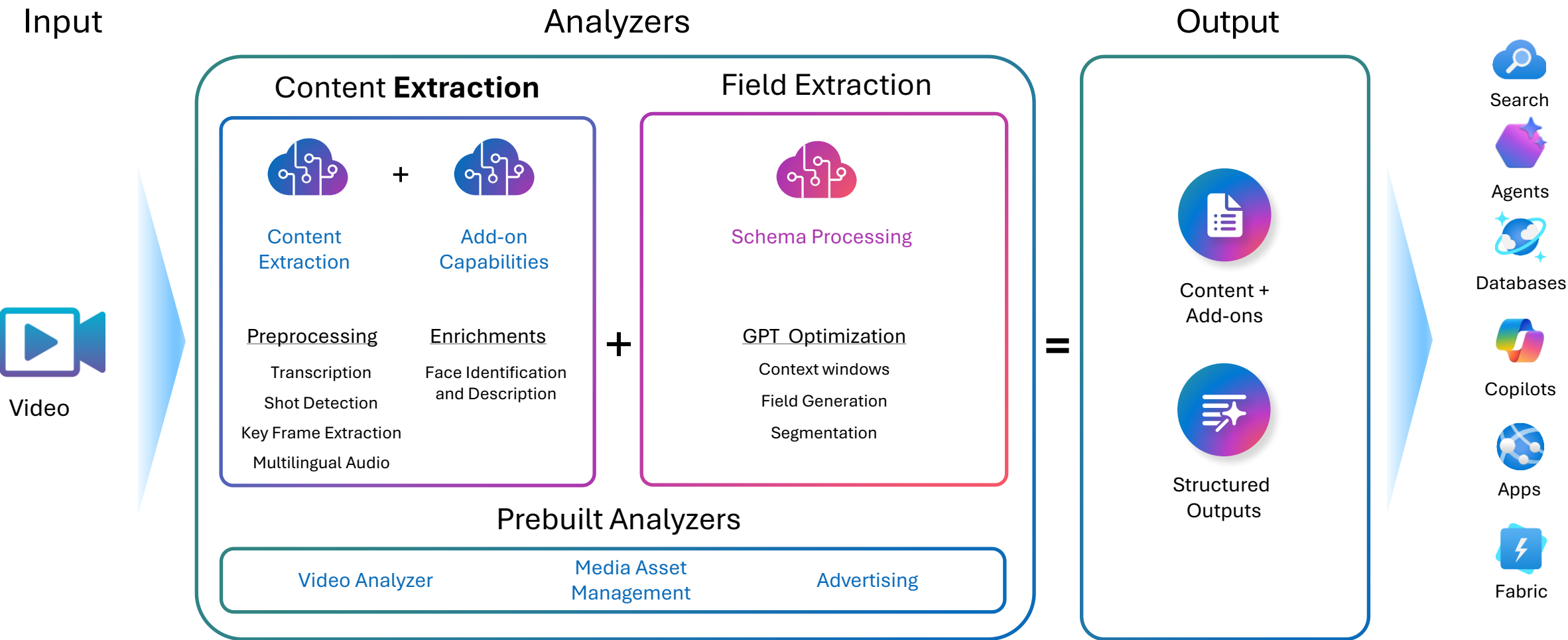


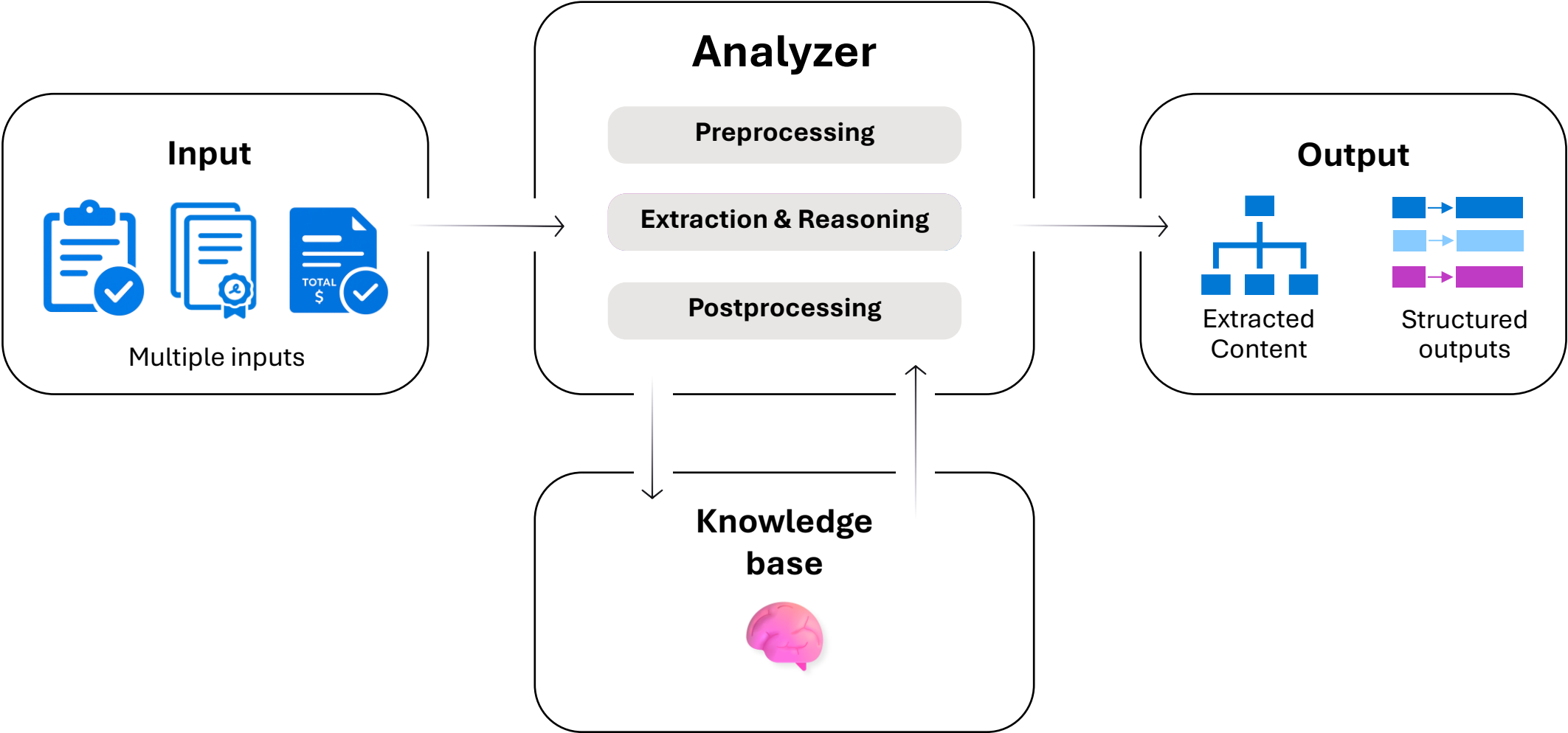
Image Capabilities



Video Capabilities



Pro Analyzers



Azure Content understanding

Standard mode vs Pro mode

Feature	Standard mode	Pro mode
 Large documents	✓	✓
 Field mode	✓	✓
 Extract, classify, and generate fields	✓	✓
 Grounding and confidence scores	✓	✗
 Input document type	Documents, images, video, audio	Documents
 Max fields	100	100
 Multiple input document processing	✗	✓
 Reference dataset integration	✗	✓
 Multi-step reasoning	✗	✓

Pro mode is a preview feature available only in the preview API version (2025-05-01-preview).

Agenda

- Présentation Azure Content Understanding
- **Démonstrations**
- Azure Document Intelligence vs Azure Content Understanding
- Ressources et documentation





Get started with Content Understanding

Content Understanding is an Azure AI service that helps you analyze and extract insights from diverse content types—documents, images, audio, and video. With prebuilt analyzers and a no-code approach, you can quickly unlock structured data or create custom solutions tailored to your business needs. [learn more](#) .



Try common prebuilt analyzers on your data

-  **Layout**
Extracts tables, checkboxes, and text layout from forms and documents.... →
-  **Document fields**
Extracts key-value pairs including structured data from any form or... →
-  **Invoice**
Extracts invoice number, customer and vendor info, billing and shipping... →
-  **Id document**
Extracts personal information from passports and ID cards, including name,... →
-  **US tax**
Classifies and extracts data from a wide range of US tax forms, including W... →
-  **Call center**
Analyzes call recordings to extract topics, sentiment, speaker roles, and call... →

[Explore all prebuilt analyzers](#) >

Recent custom projects

[View all custom projects](#)

[+ Create custom project](#)

Project name	Description	Kind	Version	Last visited on
video	Video analysis	Extract	2025-11-01	Mar 09, 2026, 04:17 PM
router	Documents routing	Classify	2025-11-01	Mar 03, 2026, 02:11 PM
boarding pass	Boarding pass demo	Extract	2025-11-01	Dec 04, 2025, 05:10 PM
call center	Call center analysis	Extract	2025-11-01	Dec 04, 2025, 02:59 PM

Learning Resources

 [Your privacy choices](#)

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Manage projects and analyzers for your custom experience

Create, configure, and fine-tune projects or analyzers to deliver solutions tailored to your business needs.

 **Project list**  Analyzer list

 Create

 Refresh

 Search

Current resource

aifoundry-sr-rg-resource-4049 

Project name	Description	Kind	Version	Created on 	
router	Documents routing	Classify	2025-11-01	Dec 05, 2025, 10:00 AM	 
boarding pass	Boarding pass demo	Extract	2025-11-01	Dec 04, 2025, 05:03 PM	 
video	Video analysis	Extract	2025-11-01	Dec 04, 2025, 03:00 PM	 
call center	Call center analysis	Extract	2025-11-01	Dec 04, 2025, 02:56 PM	 

Manage projects and analyzers for your custom experience

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 **Project list**  Analyzer list

 **Create**  Refresh

 Search

Current resource aifoundry-sr-rg-resource-4049 

Project name	Description	Kind	Version	Created on ↓		
IDS		Extract	2025-11-01	Mar 09, 2026, 04:32 PM		
router	Documents routing	Classify	2025-11-01	Dec 05, 2025, 10:00 AM		
boarding pass	Boarding pass demo	Extract	2025-11-01	Dec 04, 2025, 05:03 PM		
video	Video analysis	Extract	2025-11-01	Dec 04, 2025, 03:00 PM		
call center	Call center analysis	Extract	2025-11-01	Dec 04, 2025, 02:56 PM		

Try the new Microsoft Foundry experience

Build and deploy your AI solutions faster, safer, and with the reliability you expect. [Learn more](#)

Start building →



Overview

Model catalog

Playgrounds

Build and
customize

Agents

Templates

Fine-tuning

Content
UnderstandingObserve and
optimizeTracing PREVIEW

Monitoring

Protect and govern

Evaluation PREVIEWGuardrails +
controlsRisks +
alerts PREVIEWGovernance PREVIEW

Azure OpenAI

Stored completions

Batch jobs

My assets

Content Understanding

Overview

Try it out

Custom task

Analyzer list

+ Create

Refresh

Embark on your first Content Understanding journey by simply clicking the 'Create' button.

Run analysisBuild analyzer

Change template

File

Drag & drop files here or
[Browse for files](#)

audio.mp3

audio.mp3

00:0000:32

Content

Audio: 00:00.000 => 00:32.183

Transcript

WEBVTT

00:00.080 --> 00:00.640<v Agent 1>Good day.

00:00.880 --> 00:02.240<v Agent 1>Welcome to Contoso.

00:02.560 --> 00:03.760<v Agent 1>My name is John Doe.

00:03.920 --> 00:05.120<v Agent 1>How can I help you today?

00:05.440 --> 00:06.320<v Customer 1>Yes, good day.

00:06.640 --> 00:08.160<v Customer 1>My name is Maria Smith.

SchemaSettingsTest

Add new fieldImportExport

Field Name	Field Description	Value Type ⓘ	Method ⓘ
Summary	A one-paragraph summary	String	Generate
sentiment	Global sentiment of the discussi...	String	Generate
language	Language of the discussion. Use ...	String	Generate
agent	List the main topics mentioned ...	String	Generate
customer	List the main topics mentioned ...	String	Generate
Click to add new field			

SaveDiscard

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 **Project list**  Analyzer list

 Create

 Refresh

 Search

Current resource

aifoundry-sr-rg-resource-4049 

Project name	Description	Kind	Version	Created on 	
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video	Video analysis	Extract	2025-11-01	Dec 04, 2025, 03:00 PM	 
call center	Call center analysis	Extract	2025-11-01	Dec 04, 2025, 02:56 PM	 

← video

 Run analysis

 Build analyzer

 Change template

File


Drag & drop files here or
[Browse for files](#)




FlightSimulator.mp4

FlightSimulator.mp4





00:00

00:43

Content

Video: 00:00.000 => 00:43.866

Width: 1080 Height: 608

Transcript

WEBVTT

00:01.350 -> 00:05.610

Schema

Settings

Test

Fields

Result

⚠ Content of category `Violence` detected with severity `Low`. Please use the field content with caution.

○ summary

This video showcases a collaboration between Flight Simulator and Microsoft Azure AI, focusing on advancements in neural Text-to-Speech (TTS) technology. It highlights the importance of high-quality data for generating natural-sounding voices, the use of a universal TTS model trained on 3,000 hours of data, and the benefits of Microsoft Cognitive Services for achieving high fidelity and human-like audio. The visuals include aerial shots of landscapes, aircraft, data centers, and airport operations, emphasizing the integration of AI in realistic flight simulation experiences.

○ products

Flight Simulator, Microsoft Azure AI, Microsoft Cognitive Services

○ companies

Microsoft

Check token usage in JSON (Result tab) and learn about pricing [here](#) .

 Your privacy choices

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Azure Content Understanding - Audio Processing Showcase

Speech Intelligence · Diarization · Call Center Analytics · API 2025-11-01 (GA)

In [1]:

```
import io
import json
import matplotlib.gridspec as gridspec
import matplotlib.patches as mpatches
import matplotlib.pyplot as plt
import os
import numpy as np
import pandas as pd
import re
import requests
import sys
import time
import urllib.request
import wave

from azure.identity import DefaultAzureCredential
from collections import Counter
from datetime import datetime
from dotenv import load_dotenv
from IPython.display import Audio, display, HTML, Markdown
```

I

In [2]:

```
sys.version
```

Out[2]: '3.10.18 (main, Jun 5 2025, 13:14:17) [GCC 11.2.0]'

In [3]:

```
load_dotenv("azure.env")
```


Agenda

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Azure Document Intelligence vs Content Understanding

Capability	Document Intelligence	Content Understanding
Focus	Document processing	Generative and reasoning based content understanding
Use cases	Structured/semi structured	Multi modal
Effort	Labeled data for custom models	Zero shot extraction with schema definition
Pricing Model	Per page (fixed rates)	Per page + token-based (variable)
Cost Predictability	★★★★★ — Fixed per-page pricing	★★★★☆ — Varies with schema complexity & model
Document OCR (Read)	☑ Available	☑ Available
Layout Analysis	☑ Available	☑ Available
Prebuilt Invoice/Receipt	☑ Available	☑ Available
Custom Field Extraction	☑ Available	☑ Available
Image Processing	✗ Not supported natively	☑ Free extraction + LLM analysis
Audio Processing	✗ Not supported	☑ \$0.36/min + LLM tokens
Video Processing	✗ Not supported	☑ \$1.00/min + LLM tokens
Multimodal (Doc+Image+Audio+Video)	✗ Documents only	☑ Unified pipeline
LLM Model Choice	✗ No choice	☑ GPT-4o-mini, GPT-4.1, etc.
Confidence Scores	☑ Available	☑ Optional (increases token cost ~2x)
Source Grounding	✗ Limited	☑ Shows where data was extracted
Segmentation / Categorization	✗ Classification only (\$3/1K)	☑ AI-powered segmentation & routing
Reasoning capabilities	✗	☑ Available
Multi file processing	✗	☑ Available
Latency	Low	Medium
Free Tier	☑ 500 pages/month	✗ No free tier
Commitment Tiers	☑ Available for volume discounts	✗ Pay-as-you-go only
Maturity	★★★★★ — GA, battle-tested	★★★★☆ — GA since Nov 2025
Best For	High-volume document-only workflows with predictable pricing	Multimodal pipelines, RAG, agentic workflows, cost-optimized with mini models
Container support	☑ Available	✗ Not available

[Choose the right Azure AI tool for document processing - Foundry Tools | Microsoft Learn](#)

Cost study 1,000 Pages Scenario

Use Case	Doc Intelligence	Content Understanding
Basic OCR / Read	\$1.50	~\$1.00
Layout Extraction	\$10.00	~\$5.00 extraction (no field extraction)
Invoice Field Extraction	\$10.00	~\$6.54 (with GPT-4o-mini)
Invoice + Confidence + Source Grounding	\$10.00	~\$11–13 (2x tokens for conf.)
Custom extraction	\$30.00	~\$6–15 (schema-dependent)

Note: Content Understanding costs are estimates based on ~1,100 input + 60 output tokens/page with GPT-4o-mini. Actual costs vary by document complexity.

[Pricing for Azure Content Understanding in Foundry Tools - Foundry Tools | Microsoft Learn](#)

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Documentation

- [Azure Content Understanding documentation | Microsoft Learn](#)
- [Quickstart: Azure Content Understanding in Foundry Tools REST APIs - Foundry Tools | Microsoft Learn](#)
- [Service quotas and limits - Content Understanding - Foundry Tools | Microsoft Learn](#)
- [Content Understanding region and language support - Foundry Tools | Microsoft Learn](#)
- [Pricing - Azure Content Understanding in Foundry Tools | Microsoft Azure](#)
- <https://github.com/Azure-Samples/azure-ai-content-understanding-with-fabric>

Multimodal data processing with Azure AI Content Understanding

MULTIMODALITY USING AZURE AI

Azure AI Content Understanding: How to Get Started

with Vinod Kurpad

07:41

EPISODE

Azure AI Content Understanding: How to Get Started

Apr 1, 2025

Azure

MULTIMODALITY USING AZURE AI

Post-call analytics using Azure AI Content Understanding

with Jan Görgen

06:59

EPISODE

Post-call analytics using Azure AI Content Understanding

Apr 1, 2025

Azure

MULTIMODALITY USING AZURE AI

RAG on Video using Azure AI Content Understanding

with Joe Filcik

06:20

EPISODE

RAG on Video using Azure AI Content Understanding

Apr 1, 2025

Azure

Article Medium et GitHub Repo



<https://aka.ms/azure-cu-article>

Mastering Azure Content Understanding: A Complete Hands-On Guide

Transform unstructured documents, videos, audio, and images into structured, actionable data using Azure's newest GA AI service



Serge Retkowsky

Follow

7 min read · Dec 3, 2025



<https://aka.ms/azure-cu-repo>

Azure Content Understanding

License **MIT** python **3.8+** Azure **Content Understanding**

A collection of Jupyter notebooks demonstrating the capabilities of **Azure Content Understanding (GA version - API 2025-11-01)**. These demos showcase how to process and analyze documents, images, videos, and audio files using generative AI to extract structured data from unstructured content.

Demos videos

The screenshot displays the Azure AI Foundry interface. At the top, there's a YouTube logo and a search bar labeled 'Rechercher'. Below this, the 'Azure AI Foundry' header is visible, followed by the 'Content Understanding' section. A 'Create a new task' dialog is open, showing a form to create a new task. The form includes fields for 'Task name' (with 'Invoice analysis' entered) and 'Description' (with 'Analyse the' entered). Below these fields, there are two main options: 'Single-file content analysis' (marked as 'Recommended') and 'Multi-file content analysis' (marked as 'Preview'). The 'Single-file content analysis' option includes a description: 'Extract structured output from unstructured data using a custom-defined schema. Recommended for: most use cases building content analyzer.' The 'Multi-file content analysis' option includes a description: 'Generate insights across multiple files using contextual reference knowledge. Recommended for: complex use cases require reasoning processing.' To the right of the dialog, there's a table with columns for 'Created On' and a list of tasks. The sidebar on the left shows various navigation options like 'Overview', 'Model catalog', 'Playgrounds', 'AI Services', 'Build and customize', 'Agents', 'Templates', 'Fine-tuning', 'Content Understanding', 'Prompt flow', 'Observe and optimize', 'Tracing', 'Monitoring', 'Protect and govern', 'Evaluation', and 'Guardrails + controls'.

Azure Content Understanding
Serge Analytics - 1/5

- Azure Content Understanding Document Processing (2:16) Serge Analytics
- Azure Content Understanding Image Processing (1:45) Serge Analytics
- Azure Content Understanding Pro mode (7:18) Serge Analytics
- Azure Content Understanding Video Processing (2:06) Serge Analytics
- Azure Content Understanding Audio Processing (3:13) Serge Analytics

[Azure Content Understanding - YouTube](#)

This deck is available here

<https://aka.ms/azure-acu-pdf>

Global AI community



Global AI Community Virtuelle Francophone

The biggest AI community around the globe - communauté IA francophone, surtout virtuelle, parfois en présentiel

Prochain évènement



Présentation des accélérateurs Citadel et Spec2Cloud

Citadel et Spec2Cloud sont deux accélérateurs Microsoft IA. Citadel fournit une **gouvernance centralisée, la conformité et l'observabilité pour déployer des agents IA sécurisés et contrôlés à l'échelle**. Spec2Cloud accélère le **développement d'applications et d'agents IA via des modèles de spécifications**.

Ensemble, ils résolvent le dilemme vitesse vs contrôle : Spec2Cloud vous fait développer vite, Citadel vous assure de le faire en sécurité et conformité. Ces solutions offrent une base pour **transformer rapidement les idées en agents IA gouvernés et production-ready**.



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Alibek JAKUPOV

France AI Global Black Belt
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Nicolas ROBERT

Profession
@Cellenza
AI MVP

LinkedIn Live

jeudi

19/03/2026

à partir de 13h

https://lnkd.in/eUe_bBrz

Replays Global AI Community



<https://www.linkedin.com/company/global-ai-community-france/posts/?feedView=all>

Merci